

Detecting Man-in-the-Middle (MitM) Attacks in IoT Environments using the XGBoost Algorithm

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The growth of Internet of Things (IoT) also increases the risk of cyber attacks, among which Man-in-the-Middle (MitM) attacks are implemented that intercept and manipulate traffic on low-powered IoT devices. The low-powered IoT devices generally are based on lightweight protocols such as MQTT and CoAP, and have a limited lifespan due to their minimal encryption and authentication. The presented study proposes a detection method for MitM attack by XGBoost using the CICIoT2023 data. This method includes the preconditioning of the data, class balancing using SMOTE, training of the respective model and evaluation of it based on relevant metrics. So far the model's accuracy was 96.3%, precision 98.3%, recall 97.1%, F1-score 97.7% and ROC-AUC 0.981. SHAP analysis shows that packet size, inter-packet duration, and communication direction are the most influential features. These results prove that the XGBoost approach is effective for early detection of MitM attacks on resource-constrained IoT devices.

Keywords — IoT, Man-in-the-Middle, XGBoost, Intrusion Detection System, Machine Learning.

I. INTRODUCTION

The Internet of Things (IoT) technology has rapidly spread in many industries and has drastically changed different facets of them all. The network enables billions of Internet of Things (IoT) devices to interact with one another, exchanging data with little human mediation. The number of devices expected by experts is predicted to be over 40 billion units by 2030, with an annual growth rate of roughly 15% [1]. However, the extensive connectivity of the Internet of Things also raises new issues on cybersecurity and privacy protection, despite its many advantages in improving efficiency and process automation. Most of the IoT devices use the basic internal security mechanisms and low computing and storage capacity [2]. Even on the Internet of Things (IoT) network, the majority of communication uses unencrypted protocols or protocols with low security. In the absence of sufficient cryptographic security, approximately 98% of the data traffic for IoT is sent in plain text. Internet-of-Things (IoT) networks are extremely vulnerable to various cybersecurity attacks including Distributed Denial of Service (DDoS), botnet, and Man-in-the-Middle (MitM) attacks [3]. MitM attacks enable attackers to enter the communication exchange between two devices and subsequently intercept, modify or fake exchange data without detection [1].

MitM attacks such as DNS spoofing, HTTPS spoofing and ARP poisoning may lead to leaks of data, breakdowns in service and even physical damages to critical infrastructure

[4]. With the variety of devices (device heterogeneity) and resource constraint, the Internet of Things environment is still under the threat of MitM attack. In these scenarios, traditional security measures, such as complex encryption or two-way authentication, prove to be challenging [2]. As such, machine learning techniques have become increasingly accepted for identifying anomalies in today's network traffic as well as uncovering potentially harmful trends within IoT network traffic. According to tests carried out in the past, the performance of intrusion detection systems using machine learning techniques in IoT networks is reported to be significantly improved. When it comes to the classification of network anomalies, such models as decision trees and random forests have proven to be effective. Nevertheless, these techniques may perform poorly when deployed over large data sources or in systems where real-time processing is required [1]. Other studies utilizing regression and probabilistic models to detect MitM and DDoS attacks also exhibited low accuracy, given erratic and dynamic network traffic patterns [4]. Moreover, lightweight communication protocols (e.g., MQTT, CoAP) have very low encryption and authentication [5], which makes them more susceptible to MitM and spoofing attacks. The Gradient Boosting Extreme (XGBoost) algorithm, a method with high classification accuracy, good inference speed, and capacity to model complex nonlinear patterns in data, has emerged as a promising solution to the problem [3]. Ensemble learning methods like XGBoost are recognized as better and simpler solutions in constrained IoT [2], [3]. This differs from deep learning models, which generally require more computational resources. The goal of this study is to construct an XGBoost-based model for MitM attack detection using the CICIoT2023 dataset, which contains multiple kinds of IoT attacks including ARP spoofing, DNS spoofing etc. The aim of this project is to provide an effective and accurate intrusion detection system on which MitM attacks can be found quickly and accurately, and applicable to device in resource poor IoT environment.

II. RELATED WORK

A. Previous Research

A lot of previous work has used a variety of machine learning (ML) techniques to detect MitM attacks and other cyber threats in the IoT. Sivasankari and Kamalakkannan (2022) [6] proposed a regression model and have achieved an accuracy of 87.3% detecting MitM attacks on Internet of Things networks. However, since it does not handle very large and complex datasets, this model can only detect network traffic anomalies, which needs further development. Furthermore, Panda et al. (2022) studied various machine

learning algorithms to better facilitate intrusion detection in Internet of Things (IoT) networks. Through classifying potentially dangerous traffic on the Internet of Things network, Panda et al. (2022) performed further study which examined several ML methods. These algorithms were Random Forest (RF), Decision Tree (DT) and Support Vector Machine (SVM). According to their research, Random Forest achieved best accuracy in our study (91.5% prediction), which was superior to the Decision Tree (88.9%) and SVM (85.7%). However, model performance deteriorated in real time, since computation-intensive application, had to achieve a low computational cost for real-time processing [7]. Research by Çelik et al. (2023) also showed MitM attacks can compromise encrypted communication channels of the MQTT protocol by the MQTT protocol, it was discovered out the session hijacking mechanism to compromise encrypted communication channels. Nonetheless, ML methods can still be adopted [8], which is an opportunity, at this time, because no automatic classification model is developed. A recent study by Adewole et al. (2025) have highlighted various intrusions in IoT networks by using the XGBoost method. It had an accuracy of 96.8% on the same dataset, superior to Decision Tree (89.2%) and Random Forest (93.5%) [9]. In addition, to improve interpretability of the model, Hosain et al. (2025) merged XGBoost with Explainable AI (XAI) techniques, SHAP (Shapley Additive Explanations) and LIME, achieving an accuracy score of 97.4% [10]. The results found in this study indicate that XGBoost algorithm has more accuracy and computational efficiency compared to conventional algorithms with accuracy and the computation efficiency. However, it is uncommon to be used by these authors to identify man-in-the-middle attacks on IoT networks, especially with CICIOT2023 dataset. More adaptive detection model should be formulated in this study.

B. Internet of Things (IoT)

An Internet-connected physical network of devices that are able to communicate with one another and share data. This network is known as the IoT. This technology has been utilized in sectors such as industrial automation, healthcare, and smart cities for enhancing data-driven decision making and operational efficiency [11], [12]. The Internet of Things ecosystem contains multiple benefits, but also suffers from cybersecurity problems. This is largely because most IoT devices have limited computing capacity, memory, and resources [12], [13]. As a result of these constraints, poor authentication protocols, poor encryption protocols, and failure to manage security patches adequately are prevalent. Hence there is a higher chance of cyberattacks such as DDoS, Botnet, MitM against IoT devices [11], [14].

C. Man-in-the-Middle (MitM) Attack

The type of this kind of “man-in-the-middle” attack is when an attacker stealthily places himself between two different entities which are communicating to intercept, monitor or otherwise distort the data being sent without the knowledge of either entity (this attacks as it happens). Attacks like these, [11], [15], can have an adverse impact on the security and integrity of the data stored in IoT systems. Some of the most prevalent MitM methods are known as HTTPS breaches, which can redirect or change communication traffic between devices [11], [16]. These attacks have a range of

consequences such as theft, service denial, and widespread penetration of IoT devices, especially in the clinical and industrial automation industries [12], [14]. Due to such more complex and sophisticated attacks, anomaly detection systems built based on machine learning, now constitute one of the most potent measures in the last decade. As the level of sophistication and complexity of such attacks has been expanding over the years, the design of intrusion detection systems based on machine learning has rapidly become one of the most significant protection measures that can be implemented. This system automatically analyses communication patterns and detects anomalies in Internet of Things network traffic [11], [16].

D. Extreme Gradient Boosting (XGBoost)

An algorithm for ensemble learning that is based on gradient boosting is called XGBoost. It combines a number of weak learners usually decision trees to create a better prediction model. During the training process, XGBoost repeatedly adds new trees to improve the error or residual error of the previous trees. This gradually improves the model's performance over time.

$$\mathcal{L}(\phi) = \sum_{i=1}^n l(y_i, \hat{y}_i^{t-1} + f_{t(x_i)}) + \Omega(f_t) \quad (1)$$

Description:

$\mathcal{L}(\phi)$ = overall objective (loss) function

$L(y_i, \hat{y}_i)$ = The loss function is used to evaluate the discrepancy between the actual value (y_i) and the anticipated value ($\hat{y}_i$) produced by the model.

$f_t(x_i)$ = prediction of the t-th tree, for instance (x_i)

$\Omega(f_t)$ = regularization term used to control model complexity

$$\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \sum_j w_j^2 \quad (2)$$

Description:

$\Omega(f_t)$ = regularization term that controls the model's complexity

γT = penalty term corresponding to the number of leaves (T)

λ = L2 regularization coefficient

w_j = weight of the j-th leaf node

E. Evaluation Metrics

Some of the metrics used in network security research and machine learning-based classification systems include precision, accuracy, recall, F1-score, and ROC-AUC (Receiver Operating Characteristic – Area Under Curve). Several studies, such as those conducted by Zhang et al. (2023), Adewole et al. (2025), and Ali & Al-Sharafi (2025), have used these metrics to determine the efficiency of models in detecting anomalies within IoT networks and classifying traffic as either normal or malicious. Accuracy is expressed as the ratio of the number of correct predictions with respect to the total volume of test data applied to the model. Zhang et al. (2023) used this testing technique in their research project for machine learning intrusion detection models in IoT environments. Precision refers to the model's ability to identify attacks that have occurred, while recall is the model's ability to detect all actual attacks. The F1 score shows the

tradeoff between precision and recall. It is calculated as the harmonic mean of precision and recall. The ROC–AUC metric indicates the classifier's ability to differentiate between normal and malicious traffic. This high ROC–AUC value indicates that the model is successfully and consistently capturing the unusual communication trend of the Internet of Things (IoT) network.

1) *Accuracy*

Accuracy quantifies the ratio of correct predictions relative to the total dataset and is expressed as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

2) *Precision*

The formula for precision indicates the number of predictions that are genuinely correct attacks:

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

3) *Recall*

Recall evaluates the model's capability to identify all actual attacks that transpire.

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

4) *F1-Score*

The F1-score is the harmonic mean of precision and recall, reflecting the optimal level of model efficacy in detecting attacks while minimising classification errors.

$$F1-Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (6)$$

Description:

True Positive (TP) → signifies a MitM attack that the model successfully identified.

True Negative (TN) → denotes regular traffic accurately classified as benign.

False Positive (FP) → denotes legitimate traffic erroneously identified as an attack.

False Negative (FN) → denotes MitM assaults that the system failed to detect.

5) *ROC–AUC (Receiver Operating Characteristic – Area Under Curve)*

The ROC–AUC measure assesses the model's capacity to differentiate between benign and

malicious traffic. An elevated AUC value (approaching 1) signifies superior classification performance.

$$TPR = TP / (TP + FN)$$

$$FPR = FP / (FP + TN)$$

$$AUC \approx \sum (FPR_i^{+1} - FPR_i) \times (TPR_i^{+1} + TPR_i) / 2 \quad (7)$$

An elevated AUC indicates the model's robust ability to distinguish between attack and normal classes in IoT intrusion detection systems.

III. RESEARCH METHOD

This research explores the processes of data collection, pre-processing, data splitting, model training, model evaluation, and interpretation of findings using the SHAP method. Figure 1 shows the complete workflow for this research, illustrating the transition of processes from the first phase to the final phase of model evaluation.

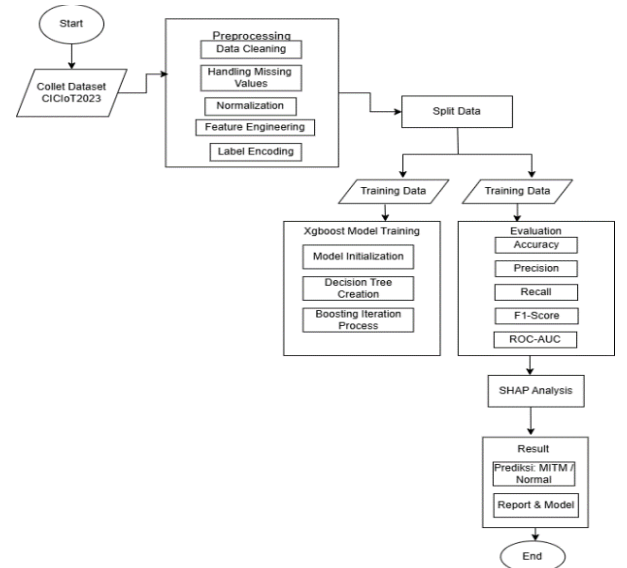


Fig. 1. Research Flow of MitM Attack Detection Using XGBoost

A. Data Collection

In this study [17], we used the CICIOT2023 dataset, one of the most contemporary thorough IoT intrusion detection datasets. The CICIOT2023 dataset consists of normal (benign) network traffic and several types of cyberattacks, such as ARP spoofing and DNS spoofing, which are forms of MitM attacks. Furthermore, this dataset includes more than 33 types of attacks and covers various types of Internet of Things devices. Such properties of this dataset make it quite suitable to be used in the training and evaluation modification processes [18].

B. Preprocessing (Data Transformation)

The data for the whole IoT network dataset has been prepared for model development by clean, normalized data set. During preprocessing, duplicates and irrelevant records were removed, missing information was efficiently filled with simple imputation, and Min–Max normalization was adopted to standardize feature distribution. Feature engineering and model selection on MitM attack patterns was

conducted for better detection [16]. We converted categorical attributes using label encoding so that they are compatible with the XGBoost algorithm. As a result, the data set became very organized and ready for training and testing.

C. Handling Class Imbalance Using SMOTE

This research addresses this problem by using synthetic minority oversampling (SMOTE) on training data, and aims at balancing the class imbalance in the CICIOT2023 dataset. This method creates synthetic samples for Man-in-the-Middle (MitM) attacks targeting minority classes by interpolating values with existing data. This method assists the model to maintain the balanced dataset volume across the normal class and the attack class so that it is not biased in the direction of the majority class. When SMOTE was implemented, the dataset became more balanced and hence the recall value was significantly increased, particularly in detection of MitM attacks.

D. Split Data (Training and Testing)

The training set constitutes eighty percent of the total data set, while the test set represents twenty percent of the total data set. This is done to provide an objective measure of how well the model generalizes [10]. Therefore the training data becomes the foundation of the parameters of the model and is used to adapt the model parameters, while the test data is used to evaluate the performance of the model in identifying Internet of Things that have not been seen before, regardless of whether it falls into the normal category [10]. To ensure that developed models have improved generalization ability without overfitting by using this methodology. The performance of the model is therefore demonstrated using the evaluation results.

E. Model Training Using XGBoost

On this study, the XGBoost algorithm was employed to build the intrusion detection model to identify MitM attacks in IoT networks. This algorithm builds an ensemble of decision trees based on XGBoost, a gradient boosting algorithm. The model's accuracy is increased by adding new trees to reduce the prediction error from the predictions made by the previous tree [15], [16]. The grid search was used to tune and optimize important parameters such as learning rate, maximum depth, number of estimators, subsamples, colsample_bytree, etc., and find optimal model parameters. We chose XGBoost algorithm due to its high predictive accuracy, scalability, and ability to handle great volumes of IoT data [16].

F. Evaluation

This evaluation step performed the model evaluation of the XGBoost, that is training the model to be able to detect MitM attacks on IoT network traffic. This overview was taken by considering certain major classifiers, i.e., precision, accuracy, recall, F1 score, and ROC-AUC. These are helpful to assess whether the model distinguishes normal traffic from a likely threat. The study's results were analyzed by way of a confusion matrix and ROC curve to communicate classification accuracy and the detection of attacks based on the model. Using this analysis, we demonstrate how well and successfully this XGBoost algorithm captures MitM attacks [10].

G. SHAP Analysis

This study applied the SHAP method to gain clarity on the contribution of each feature to the prediction outputs of the model. This approach improves reliability and clarity in the model developed illustrating the impact of every variable in this MitM attack classification [10]. Such an analysis allows us to have a measure of both model performance and also to have insight into how the XGBoost model is making decisions under the circumstances of IoT network intrusion detection. SHAP values to describe the results can help uncover the IoT traffic features that involve the model decision whether to detect an attack or not.

IV. RESULTS AND DISCUSSION

A. Experimental Results

This work adopted the CICIOT2023 dataset to train the model to identify Man-in-the-Middle attacks on the IoT network. Preparation of the data was carried out in several phases before training. The data were preprocessed through cleaning, input normalization, and split (80% training and 20% test). The XGBoost method was chosen for implementation during the model process because it is capable of handling large, wide, and unbalanced data. Experimental results are summarized in Figure 4, depicting the confusion matrix of the XGBoost model and classification performance on IoT network traffic.

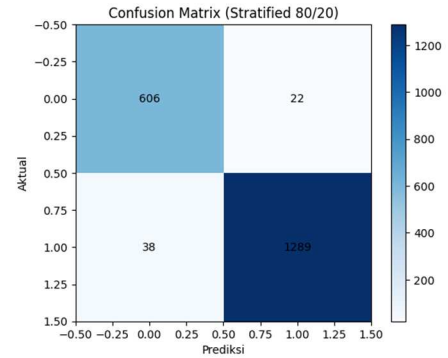


Fig. 2. Confusion Matrix (Stratified 80/20)

The test results indicate that the XGBoost model can attain superior classification performance while preserving an ideal equilibrium between accuracy and recall metrics. The model exhibits significant discriminatory strength, allowing it to effectively differentiate between regular traffic and assaults on IoT networks, evidenced by an AUC value nearing 1.

B. Comparison with Other Algorithms

This led to a comparison of performance between two decision tree-based algorithms (e.g., decision trees (DT) and random forests (RF)) that were employed using the same parameter settings [11],[19]. Both algorithms are interpretable, perform well for classification, and are common base classifiers for intrusion detection over IoT networks. The results of performance evaluation of the three algorithms favor the viability of the approach proposed in this report for MitM threat detection while appropriately segregating IoT network traffic when it is normal.

TABLE I. COMPARISON WITH OTHER ALGORITHMS

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC
DT [11]	94.97%	99.25%	94.99%	97.07%	0.972
RF [19]	95.03%	99.68%	94.64%	97.10%	0.975
XGBoost [Our Work]	96.30%	98.30%	97.10%	97.70%	0.981

The RF model demonstrated exceptional performance, achieving an accuracy of 95.03%, precision of 99.68%, recall of 94.64%, and an F1 score of 97.10%. Conversely, the DT model demonstrated similar outcomes, achieving an accuracy of 94.97%, a precision of 99.25%, a recall of 94.99%, and an F1 score of 97.07%.

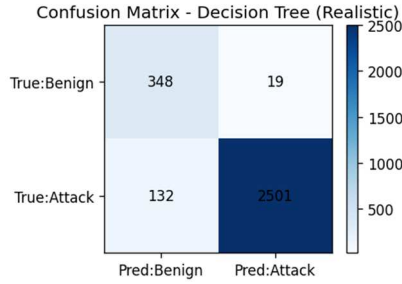


Fig. 3. Confusion Matrix – Decision Tree

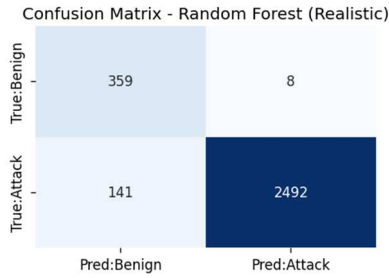


Fig. 4. Confusion Matrix – Random Forest

According to the test metrics, the results show an overall outperformance by the XGBoost model over RF and DT. XGBoost adopts gradient boosting methodology that enhances the performance of the model by gradually learning to fix the past errors in classification. In addition, regularization of (L1 and L2) becomes important for complexity of the model. L1 and L2 regularization reduces the risks of overfitting and stabilizes the predicted outcomes. Compared to other tree based algorithms, XGBoost also shows strong ability to detect MitM attack patterns when utilizing L1 and L2 regularization techniques.

C. Performance Evaluation Metrics

The precision of the XGBoost model (98.3% and recall 97.1%) and accuracy of XGBoost (96.3%) were observed. The value of the F1-score (97.7%) showed a good balance between the precision level and recall value, and for ROC-AUC (0.981) we saw that the model can correctly discriminate between the normal and malicious data. The performance shows that the XGBoost technique provides sufficient stability and reliability in detecting MitM attacks in the Internet of Things (IoT) network scenario.

D. Model Interpretability Analysis (SHAP)

To achieve better model transparency, detailed analysis was performed to evaluate the contribution of each feature on prediction outcomes with SHAP. Several of the most important features, like `network_packets_all_count`, `network_tcp-flags-ack_count`, `network_time-delta_max`, `network_ips_src_count`, and `network_ports_all_count`, point to abnormal communication behavior, high number of packets, or irregular time frames between packets, and are considered warning signs of an attack. An SHAP Feature Importance plot – to visualize each of the features' contribution to the classification – was also shown in Figures 5 and 6. Positive SHAP values indicate an increase in the likelihood of an event being classified as an Attack, while negative SHAP values signify that traffic is more likely to be Normal. The results of this analysis are consistent with our results, where high positive SHAP values (e.g. `network_packets_all_count` and `network_tcp-flags-ack_count`) are more likely to detect malicious activity, while factors with high negative SHAP values (e.g. `network_protocols_all_count` and `network_mss_avg`) match normal traffic. The SHAP analysis results provide insight into the established XGBoost model with a high degree of interpretability for normal communication patterns and IoT network anomalies overall.

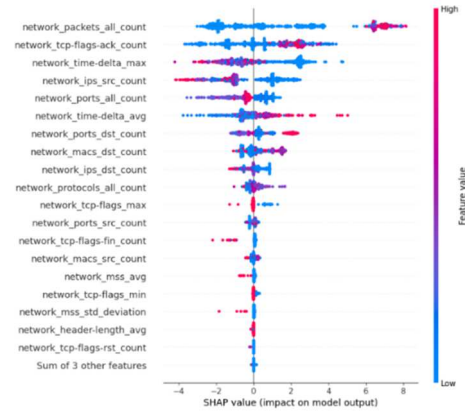


Fig. 5. Shap Value

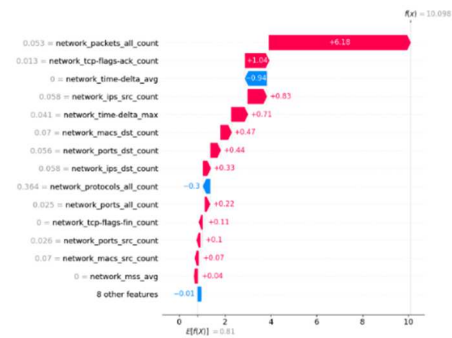


Fig. 6. Shap Value

This study employs the SHAP approach to identify the factors that most significantly impact the MitM attack detection procedure. This investigation is done to gain a better understanding of how the XGBoost model works. As illustrated in Figures 5 and 6, the characteristics `network_packets_all_count`, `network_tcp-flags-ack_count`, and `network_time-delta_max` exhibit the greatest SHAP values. This result signifies that these qualities are

substantially accountable for the categorisation of malicious traffic, sometimes referred to as hazardous traffic. A primary indicator of MitM attack behaviour on the IoT network is the volume of network packets, TCP flag activity, and anomalies in the interpacket timing. The study's results indicate that the XGBoost model exhibits both excellent detection accuracy and the capability to elucidate the network properties linked to MitM attack patterns.

E. False Positive Analysis

The model generates 22 false positives from 628 benign samples and gives a false positive rate of 3.50%. This value is relatively low and consistent with the high precision of the model (98.3%). Nevertheless, false positives have to be considered because they are liable to trigger false alarms in real applications. Therefore further study is needed to confirm that the model remains stable in real conditions of the IoT network.

F. Discussion

The results show the XGBoost algorithm reveals MitM attacks in IoT more effectively than DT and RF. The combination of adaptive boosting and regularization improves the model's capacity to analyze large and imbalanced network data. The outcomes of the SHAP analysis indicate that `network_packets_all_count` and `network_time-delta_max` are two important metrics of abnormal behaviour on the IoT network. The XGBoost model provides stability, efficiency and reliability as an intrusion detection method, with an overall accuracy of 96.3% and an AUC value of 0.981. This is of particular relevance for IoT devices that have limited resources in terms of computation.

V. CONCLUSION

After completing this study, we demonstrated that XGBoost can effectively detect man-in-the-middle (MitM) attacks, especially in the Internet-of-Things (IoT) scenario. Using metrics from the CICIoT2023 dataset, we achieved a high performance, achieving 96.3% accuracy, 98.3% precision, 97.7% F1 score, and 0.981 ROC-AUC for recognition. In addition, XGBoost demonstrated discriminability, that is to say whether the traffic is benign or malicious, helping distinguish the two, and thus it is more computationally efficient than the other algorithms (DT and RF). This was especially crucial for classification characteristics and robustness against overfitting, because of its regularization and general boosting approach. We also verified the explainability of the two main attributes for predicting MitM attack activity through interpretability analysis based on SHAP and network-based features (`network_packets_all_count` and `network_time-delta_max`).

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